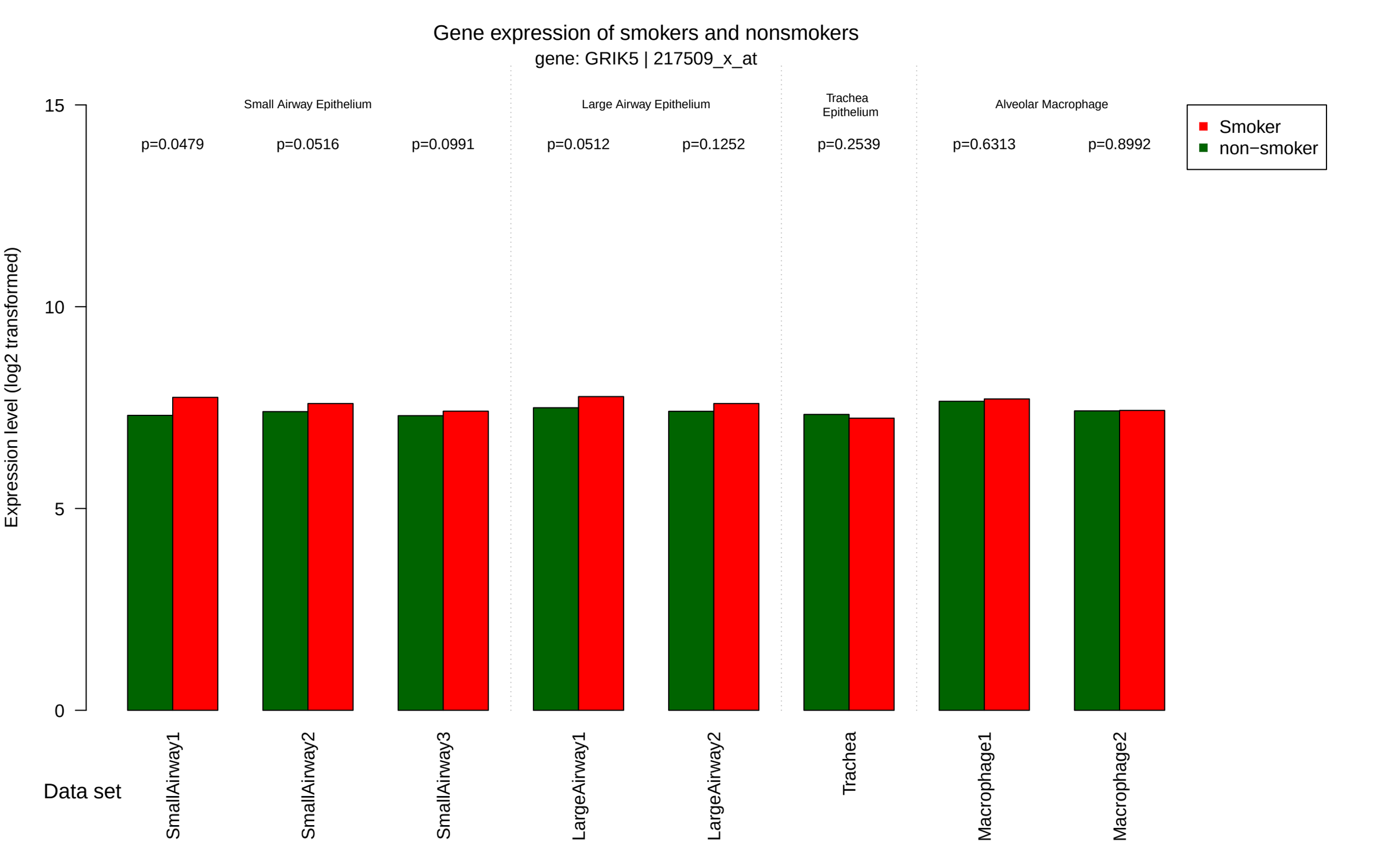
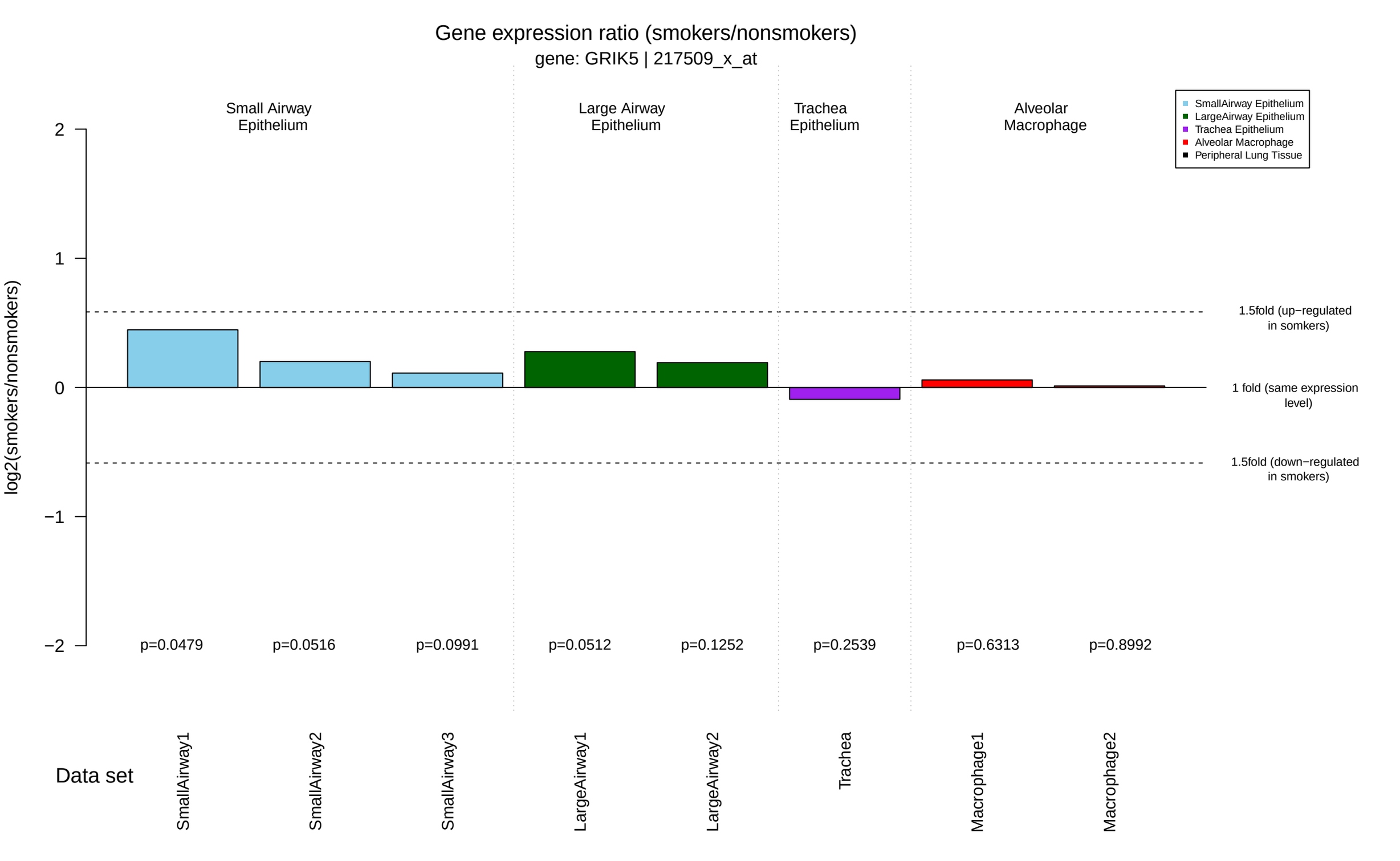
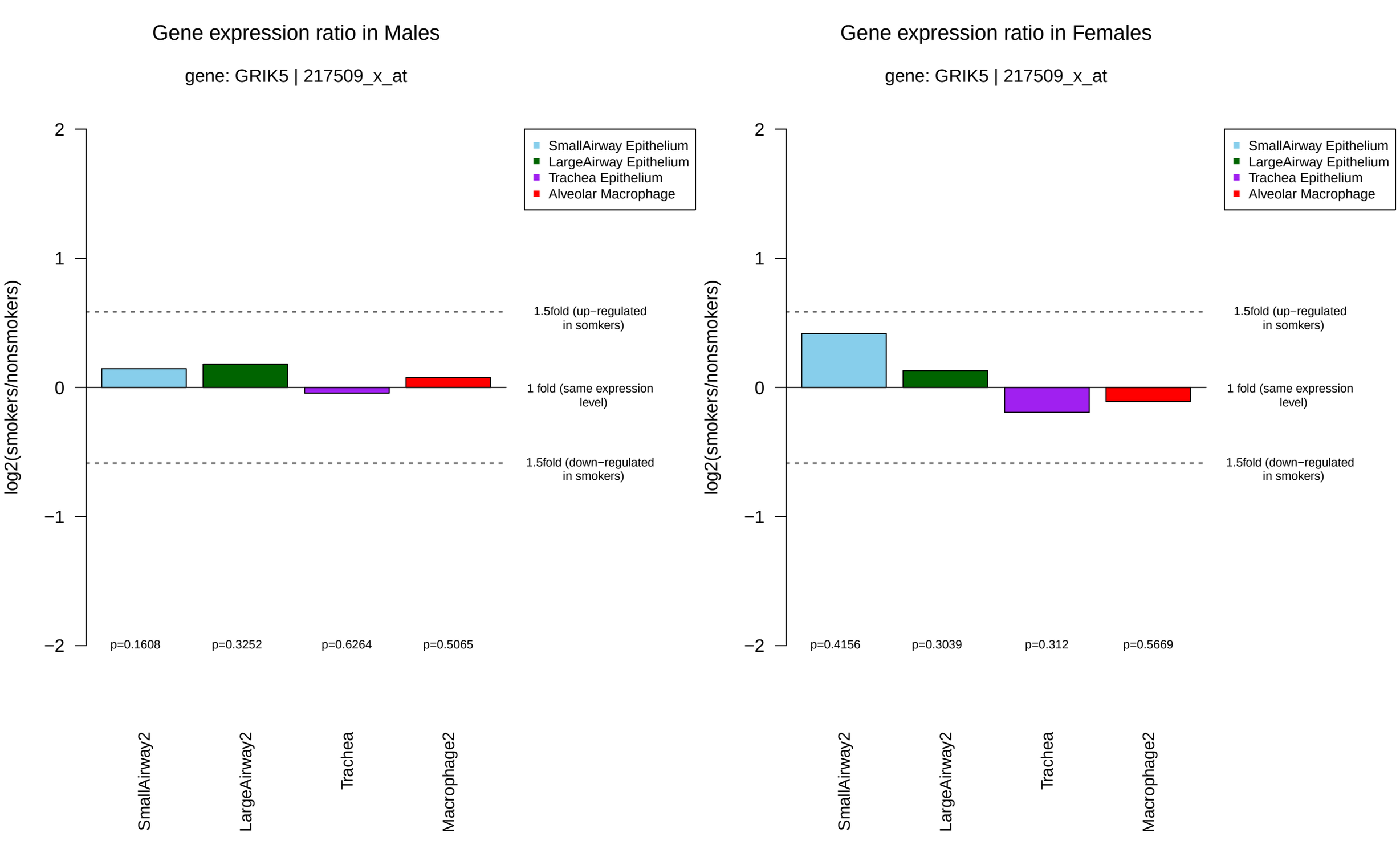
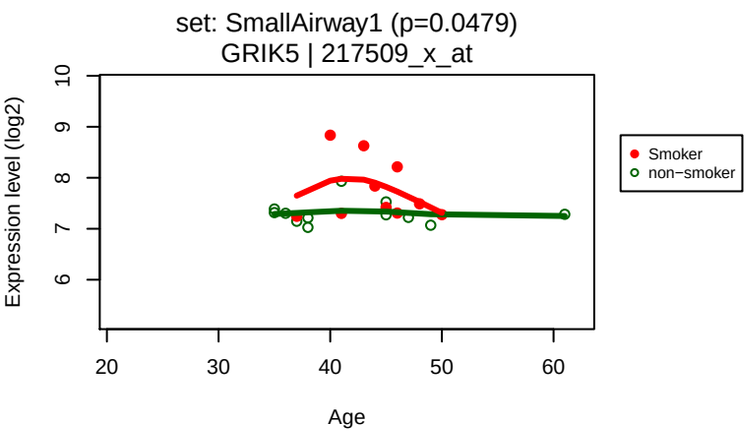
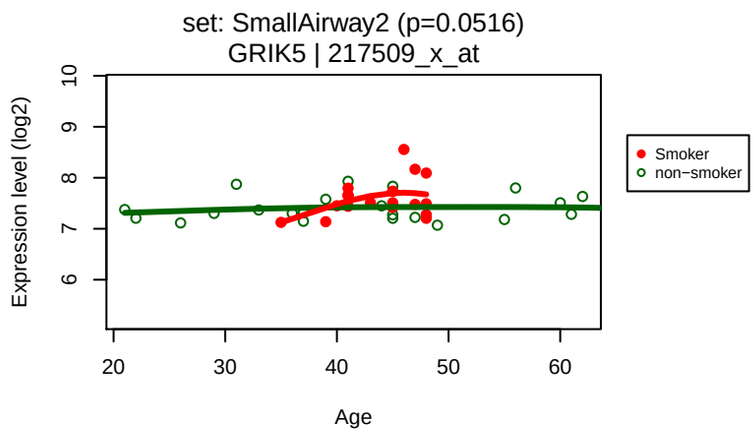
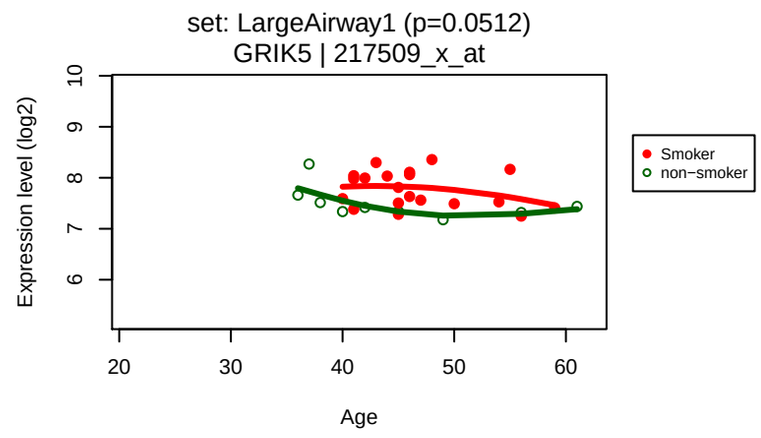

Gene expression of smokers and nonsmokers
gene: GRIK5 | 217509_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0479
p=0.0516
p=0.0991
p=0.0512
p=0.1252
p=0.2539
p=0.6313
p=0.8992
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

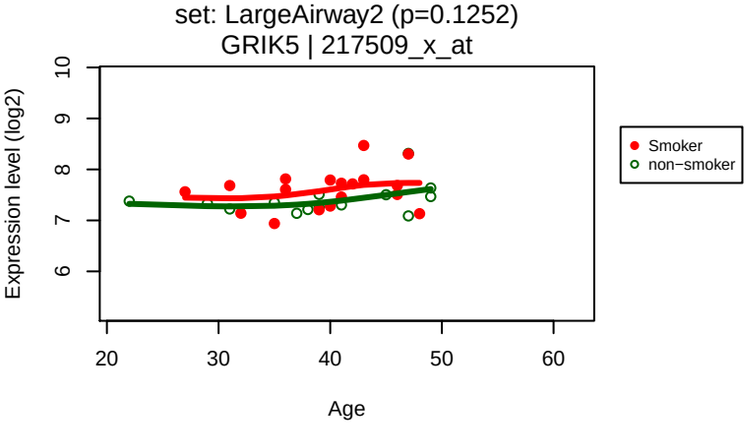

Gene expression ratio (smokers/nonsmokers)
gene: GRIK5 | 217509_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0479
p=0.0516
p=0.0991
p=0.0512
p=0.1252
p=0.2539
p=0.6313
p=0.8992
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

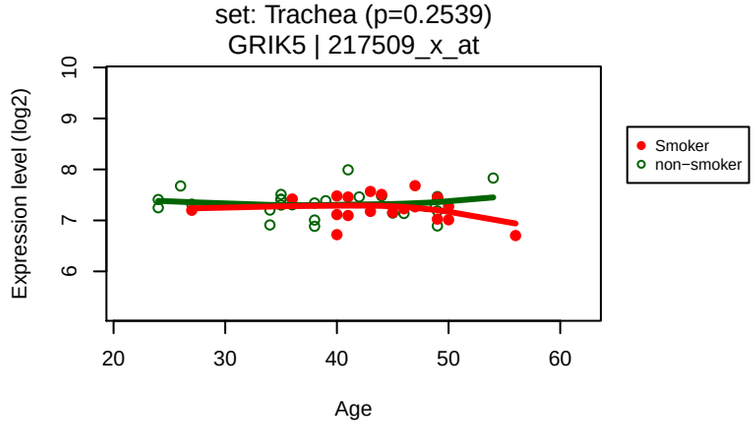

Gene expression ratio in Males
gene: GRIK5 | 217509_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1608
p=0.3252
p=0.6264
p=0.5065
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIK5 | 217509_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4156
p=0.3039
p=0.312
p=0.5669
SmallAirway2
LargeAirway2
Trachea
Macrophage2

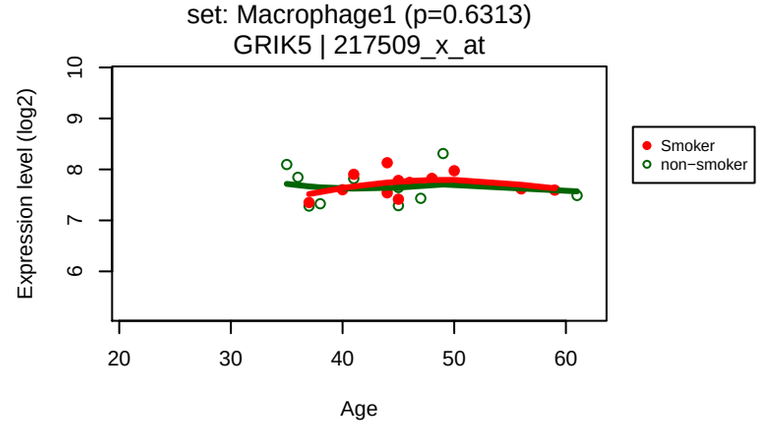

set: SmallAirway1 (p=0.0479)
GRIK5 | 217509_x_at
Expression level (log2)
Age
Smoker
non−smoker

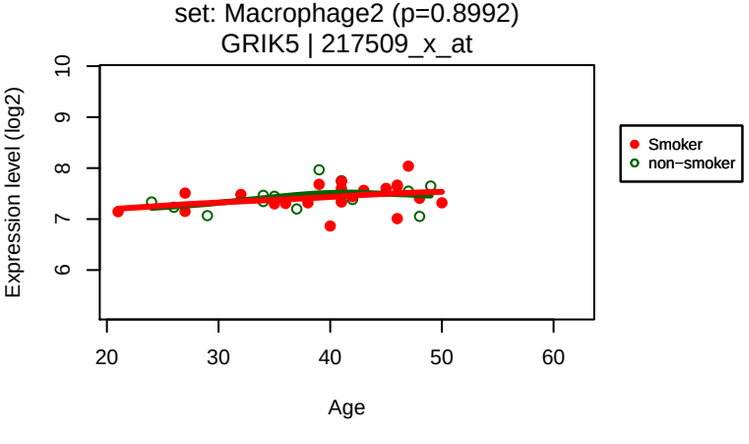

set: SmallAirway2 (p=0.0516)
GRIK5 | 217509_x_at
Expression level (log2)
Age
Smoker
non−smoker

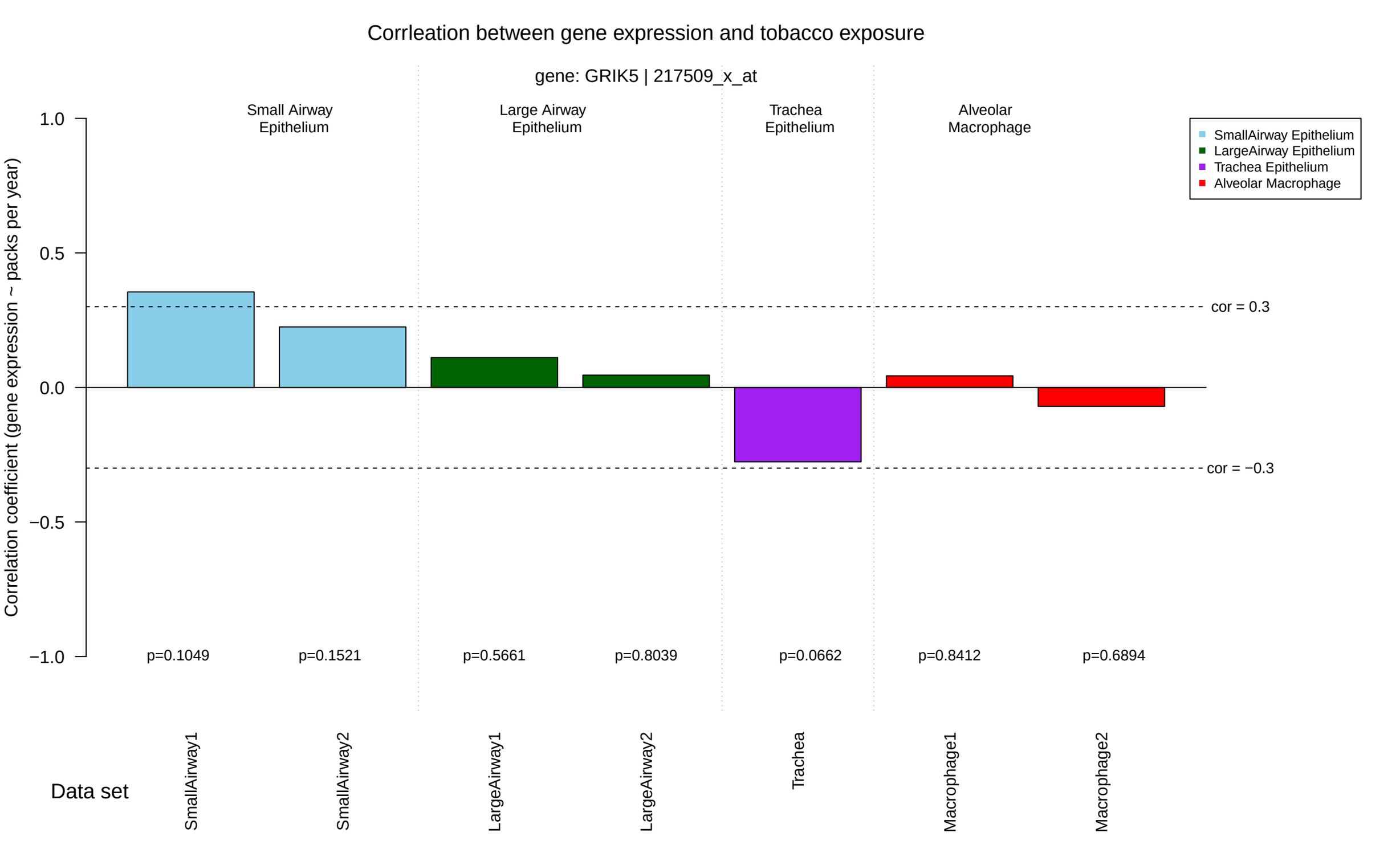

set: LargeAirway1 (p=0.0512)
GRIK5 | 217509_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1252)
GRIK5 | 217509_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2539)
GRIK5 | 217509_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6313)
GRIK5 | 217509_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8992)
GRIK5 | 217509_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIK5 | 217509_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1049
p=0.1521
p=0.5661
p=0.8039
p=0.0662
p=0.8412
p=0.6894
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2